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Final Project: Paper Selection
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Bloom, J. C., Torres, E., Pereira, S. A., Arvizu-Sanchez, L., Fontes, A. N., Joffe, H., Page, D. C., & Navarro, V. M. (2026). Progressive hypothalamic neuroinflammation in ovariectomized mice parallels aging-related transcriptomic changes in the female human hypothalamus. *JCI insight*, *11*(17), e207270. https://doi.org/10.1172/jci.insight.207270

SUMMARY

Objective

To characterize how the mouse hypothalamus changes over time after estrogen withdrawal, whether short- and long-term ovariectomy (OVX) produce distinct molecular programs, and whether those programs resemble age-related changes in the female human hypothalamus.

Bioinformatic Approach

Bulk RNA-seq of two hypothalamic regions, the posterior hypothalamus (PH, KNDy neurons) and preoptic area (POA, thermoregulation). Reads pseudoaligned with kallisto (mm39, GENCODE vM30), imported with tximport (v1.24.0), analyzed in DESeq2 (v1.36.0) with one surrogate variable from smartSVA (v0.1.3) as a covariate (FDR < 0.05). Pathways were tested with fgsea (v1.22.0, Hallmark sets), with STRING networks and BisqueRNA cell-type deconvolution. Human GTEx data were analyzed by DESeq2 across age bins, LOESS trajectories, and cross-species GSEA correlation.

Data Source

Mouse: female C57BL/6 mice, intact vs. 14 days vs. 4 months post-OVX; 33 RNA-seq samples (PH and POA) from 17 mice, 50 bp paired-end NovaSeq, GEO GSE288245.

Physiology: (LH and core temperature) were measured in separate cohorts.

Human: (outside our scope) GTEx hypothalamus from 31 female donors aged 28–65, grouped into five age bins. Code is shared on GitHub (jcbloom/hypo_menopauseNavarro).

Principal Results

LH rose to a peak at 2 months post-OVX then declined; core temperature peaked at 14 days and normalized by 4 months, mirroring the human menopause transition. Transcriptomic change grew over time: protein-coding DEGs vs. intact rose from 13 (PH) and 14 (POA) at 14 days to 82 (PH) and 70 (POA) at 4 months, with inflammatory pathways (TNF-α/NF-κB, interferon, IL6/JAK/STAT3, complement) most enriched and estrogen-response pathways depleted at 4 months . Immunofluorescence showed reduced NKB stores, ~50% fewer activated KNDy neurons and rising GFAP at 4 months . In humans, the 51–55 age group showed similar inflammatory enrichment and 28 DEGs vs. under-45 donors, with the strongest cross-species correlation between 4-month PH and ages 51–55.

Strengths

Temporal design with two time points and two regions; physiological, transcriptomic, and protein-level evidence; raw reads publicly available with the authors' analysis pipeline (available on GitHub), sample key, and tool versions shared; cautious framing of human data as age-related rather than menopausal.

Limitations

Stated by authors: no age-matched intact controls for the 4-month group; surgical OVX models abrupt rather than gradual menopause; GTEx lacks menopausal status and hormone data, with only 31 donors.

Identified by us: GEO sample labels (4 PH sham / 6 PH 14-day) conflict wih the analysis code (5 / 5); GEO says "sham" while the paper says "intact"; apparent unreported sample exclusions; a known batch variable is not modeled, and SV1 largely separates brain regions; PH and POA from the same mouse are modeled as independent; kallisto version and settings are not give; n = 4-6 per group limits power for individual genes. Pipeline references "LOES after Piehl, et al." (Piehl et al., 2022), which may need implementation. The paper's methods section describes only the filters used to obtain the human data from the GTEx portal, which could be problamatic; however, the Supplementary Materials provide a list of included donors/samples (n=31).

Implications and Application

Long-term OVX may better model later stages of reproductive aging than standard short-term OVX, offering a preclinical platform for menopause-related hypothalamic dysfunction. The delayed inflammatory signature suggests timing matters for interventions such as hormone therapy, and the KNDy findings relate to NK3R-targeting treatments for hot flashes.

JUSTIFICATION

Why We Selected This Article

Our group screened several recent RNA-seq papers against the project criteria and a reproducibility checklist (raw data availability, sample size, documented pipeline, shared code, and clear reported targets). We selected Bloom et al. because it best balanced feasibility, scientific interest, and opportunity for meaningful analysis.

It meets all project requirements: published in 2026; an RNA-specific line of inquiry in mouse; 33 bulk RNA-seq samples from 17 mice, including intact controls, with 10–11 samples in each pairwise comparison.

It has fully accessible data. Raw reads are public in GEO/SRA (GSE288245; 33 paired-end runs, one per sample), so we can reproduce the pipeline from FASTQ files rather than processed tables.

It documented and partially shared analysis. Tool versions are reported (tximport 1.24.0, DESeq2 1.36.0, smartSVA 0.1.3, fgsea 1.22.0), and the authors' GitHub repository includes the DESeq2 code, a sample key with animal IDs and batch, and precomputed surrogate variable values.

It has clear, quantitative targets. Protein-coding DEG counts vs. intact (PH 14d = 13, PH 4mo = 82, POA 14d = 14, POA 4mo = 70), the top gene (Gm28040), and the Hallmark GSEA heatmap (Fig. 1D) give concrete benchmarks.

It is a genuine reproducibility test. Our preliminary review found gaps worth evaluating: GEO sample labels conflict with the analysis code, kallisto's version and settings are unstated, a known batch variable is not modeled while the surrogate variable largely tracks brain region, and samples appear to have been excluded without being reported. Additionally, the “LOESS after Piehl, et al” comments will need to be investigated further as noted above.

It fits with our interests and the course. The menopause/women's health topic matches our group's interests, and the bulk RNA-seq pipeline aligns with the first half of the course.

Scope

We will reproduce the mouse analysis (Figs. 1D and 2). The human GTEx analysis is outside our scope because the paper's 5-year age bins require exact donor ages, which GTEx provides only through controlled access (dbGaP).

Preliminary Plan

To strengthen the study and ensure each member learns the full pipeline, each of us will run the complete analysis independently on the university HPC, using shared reference files and sample sheet, then compare results at each checkpoint. Code, environment files, and small outputs will be tracked in a shared GitHub repository, with large files kept off GitHub.

Phase 1 (by Oct 28 progress report): downlaod FASTQs with the SRA Toolkit; run FastQC; build a kallisto index from GENCODE vM30 transcripts (mm39); quantify all 33 samples; archive the abundance files (and file logs) to persistent storage before the 30-day purge. Resolve the sample-label conflict by mapping GEO samples to the authors' library IDs.

Phase 2 (by Nov 18 progress report): import with tximport; run DESeq2 with the published design (~SV1 + condition) using the authors' SV1 values; compare protein-coding DEG counts and top genes to Fig. 2; run fgsea on Hallmark gene sets and compare to Fig. 1D.

Phase 3 (by Dec 7 final): compare our three independent reproductions; document how unstated choices (kallisto version, strand setting, annotation source) affect results; if within project scope, test sensitivity to the batch/SV1 modeling choice; prepare the final report.

References

Bloom, J. C., Torres, E., Pereira, S. A., Arvizu-Sanchez, L., Fontes, A. N., Joffe, H., Page, D. C., & Navarro, V. M. (2026). Progressive hypothalamic neuroinflammation in ovariectomized mice parallels aging-related transcriptomic changes in the female human hypothalamus. *JCI insight*, *11*(17), e207270. https://doi.org/10.1172/jci.insight.207270

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